

In21-S4-PL2406
Environmental Modelling and Assessment

Air quality Modelling
Colombo

212339B
Nivethika. S
Department of Town & Country Planning
University of Moratuwa.

Selected study area – Colombo

The dynamic capital of Sri Lanka, Colombo, faces serious air quality issues as a result of a complicated interaction between human, climatic, and topographical variables. Colombo is a thriving metropolis on the western coast of the island country. Its varied terrain, fast urbanization, and industrial operations all contribute to the buildup of air pollution. With an emphasis on important emission sources and environmental dynamics for efficient air quality modeling and management, this research seeks to give a thorough overview of the problems with air quality in the Colombo region.

Its geographical location plays a significant role in shaping its air quality. Colombo experiences a tropical rainforest climate characterized by high temperatures and humidity throughout the year. The city's proximity to the Indian Ocean influences its weather patterns, with monsoon seasons bringing heavy rainfall from May to August and October to January. Additionally, Colombo's coastal location exposes it to sea breezes, which can transport pollutants inland. These meteorological factors contribute to the dispersion and accumulation of air pollutants in the Colombo urban area.

Major emission sources

1. **Vehicle Emissions** - Due to a high vehicle density, including automobiles, buses, lorries, and motorcyclists, Colombo confronts significant air quality concerns. Significant emissions of nitrogen oxides (NO_x), carbon monoxide (CO), particulate matter (PM), and volatile organic compounds (VOCs) are caused by the constant influx of automobiles and traffic congestion. Vehicle emissions are mostly transported via the road network, particularly during rush hour and in heavily populated regions.
2. **Building and Dust** - As a result of Colombo's rapid urbanization and construction activity, air quality problems are made worse by dust emissions. Particulate matter (PM₁₀ and PM_{2.5}) is produced at construction sites by operations including excavation, demolition, and material transportation. Open spaces and unpaved roads can add to dust pollution, particularly in windy and dry weather.
3. **Industrial Activities**: Textile mills, food processing companies, chemical factories, and other manufacturing facilities make up Colombo's industrial sector. During the production process, these sectors release pollutants such sulfur dioxide (SO₂), nitrogen oxides (NO_x), particulate matter (PM), and volatile organic compounds (VOCs). Hotspots for localized air pollution are exacerbated by industrial pollutants, especially in industrial zones and the neighborhoods that surround them.
4. **Burning Biomass**: In and around Colombo, burning garbage, burning biomass, and agricultural activities all contribute to air pollution, especially during the dry season. Carbon monoxide (CO), particulate matter (PM), and volatile organic compounds (VOCs) are among the pollutants released into the environment when agricultural leftovers like rice straw and crop stubble are burned outdoors. The informal burning of solid garbage and the cremation of household waste further worsen the urban area's air quality.

Existing air quality monitoring data attached with the submission folder. I select 2018 Colombo air quality data for this assessment purposes.

Data link - /dos/historical/Colombo/2018/Colombo_PM2.5_2018_YTD.csv

Process of API workflow

I written a code to retrieve and analyze air quality data for Colombo in 2018 using Google Collab.

1. Data Retrieval - I use the ``wget`` command to download a CSV file containing air quality data for Colombo in 2018 from the specified URL. The ``!wget`` command allows you to execute shell commands directly within the Collab notebook.
2. Data Loading - After downloading the CSV file, I use the ``pandas`` library to load the data into a Data Frame (``df``). This Data Frame contains columns such as 'Date (LT)', 'NowCast Conc.', 'Raw Conc.', etc., representing various attributes of the air quality data.
3. Data Visualization - I perform exploratory data analysis (EDA) by visualizing the air quality data. I plot the 'NowCast Conc.' and 'Raw Conc.' columns using Matplotlib to observe the trends and distribution of particulate matter (PM2.5) concentrations over time.
4. Data Preprocessing - I preprocess the Data Frame by converting the 'Date (LT)' column to datetime format and setting it as the index for time series analysis. Additionally, I drop unnecessary columns ('Site', 'Parameter', etc.) and filter out rows with negative 'Raw Conc.' values.
5. Data Analysis - I further analyze the preprocessed data by computing descriptive statistics and visualizing the distribution of particulate matter concentrations. I also resample the data to calculate daily mean concentrations for a smoother representation of trends.

Overall, my code demonstrates a basic workflow for retrieving, loading, visualizing, and analyzing air quality data for Colombo in 2018 using Python and Google Collab.

Screenshot_1

```

AirQuality_Colombo.ipynb
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[ ] import os

[ ] !wget https://dosairnowdata.org/dos/historical/Colombo/2018/Colombo_PM2.5_2018_YTD.csv -P ./data/
--2023-11-12 04:18:19-- https://dosairnowdata.org/dos/historical/Colombo/2018/Colombo_PM2.5_2018_YTD.csv
Resolving dosairnowdata.org (dosairnowdata.org)... 74.208.236.6, 2607:fcb:100f:f000::279
Connecting to dosairnowdata.org (dosairnowdata.org)[74.208.236.6]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 852964 (833K) [text/csv]
Saving to: './data/Colombo_PM2.5_2018_YTD.csv'

Colombo_PM2.5_2018_100%[=====] 832.97K  847KB/s   in 1.0s
2023-11-12 04:18:21 (847 KB/s) - './data/Colombo_PM2.5_2018_YTD.csv' saved [852964/852964]

[ ] os.listdir('./data/')

[ ] import pandas as pd
df = pd.read_csv('./data/Colombo_PM2.5_2018_YTD.csv')

[ ] df.head()

```

	Site	Parameter	Date (LT)	Year	Month	Day	Hour	NowCast Conc.	AQI	AQI Category	Raw Conc.	Conc. Unit	Duration	QC Name
0	Colombo	PM2.5 - Principal	2018-01-01 01:00 AM	2018	1	1	1	92.5	170	Unhealthy	134.0	UGM3	1 Hr	Valid
1	Colombo	PM2.5 - Principal	2018-01-01 02:00 AM	2018	1	1	2	97.8	173	Unhealthy	103.0	UGM3	1 Hr	Valid
2	Colombo	PM2.5 - Principal	2018-01-01 03:00 AM	2018	1	1	3	89.9	169	Unhealthy	82.0	UGM3	1 Hr	Valid
3	Colombo	PM2.5 - Principal	2018-01-01 04:00 AM	2018	1	1	4	72.9	160	Unhealthy	56.0	UGM3	1 Hr	Valid
4	Colombo	PM2.5 - Principal	2018-01-01 05:00 AM	2018	1	1	5	56.9	152	Unhealthy	41.0	UGM3	1 Hr	Valid

```

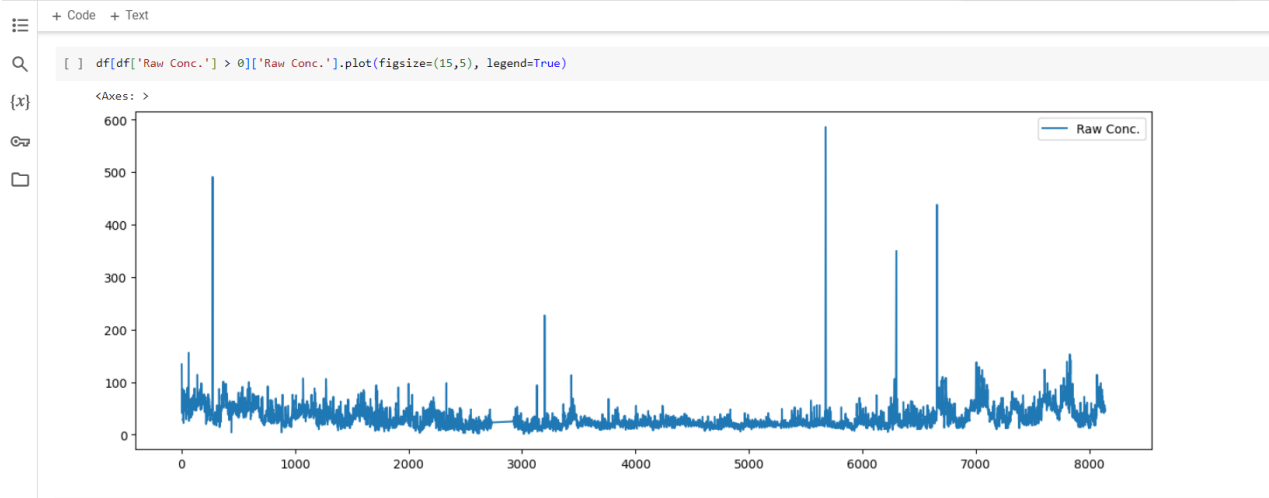
[ ] df[['NowCast Conc.', 'Raw Conc.']].plot(subplots=True, figsize=(15,10));

```

Screenshot_2



Screenshot_3



Screenshot_4

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Screenshot_5

+ Code + Text

```
[ ] df.drop(columns=['Site', 'Parameter', 'Year', 'Month', 'Day', 'Hour', 'Conc. Unit', 'Duration'], inplace=True)
df.head()
```

Date (LT)	NowCast Conc.	AQI	AQI Category	Raw Conc.	QC Name
2018-01-01 01:00:00	92.5	170	Unhealthy	134.0	Valid
2018-01-01 02:00:00	97.8	173	Unhealthy	103.0	Valid
2018-01-01 03:00:00	89.9	169	Unhealthy	82.0	Valid
2018-01-01 04:00:00	72.9	160	Unhealthy	56.0	Valid
2018-01-01 05:00:00	56.9	152	Unhealthy	41.0	Valid

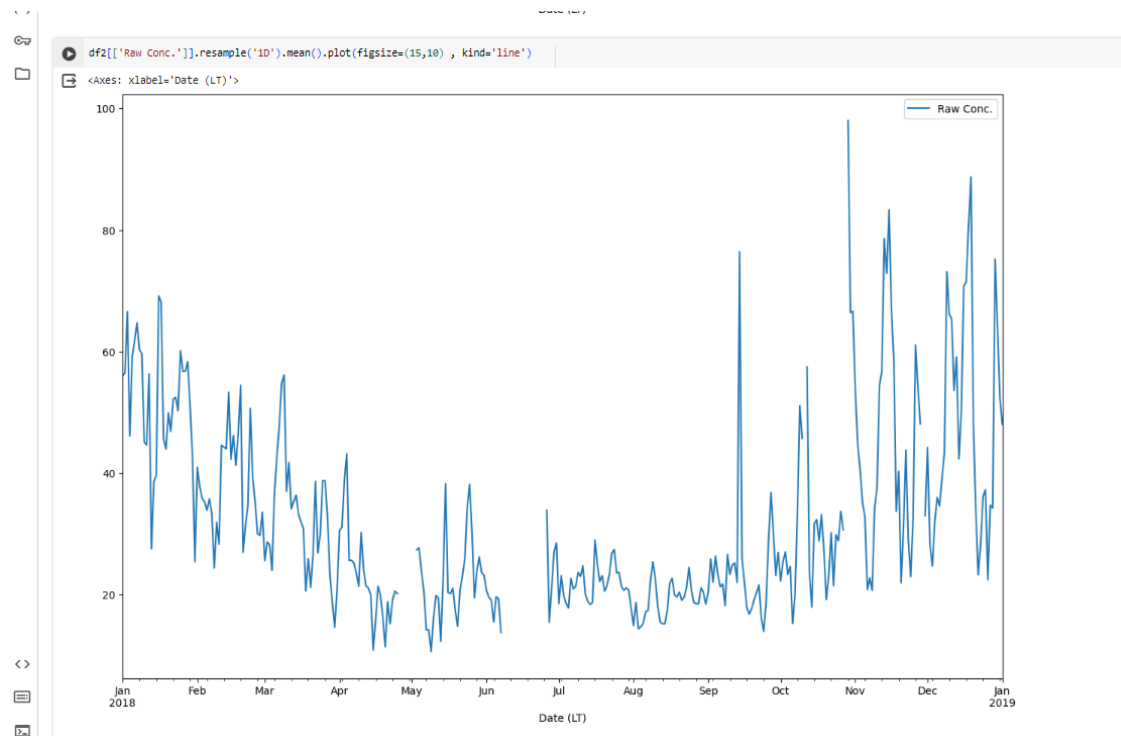
```
[ ] df2 = df[df['Raw Conc.']>=0]
df2.describe()
```

	NowCast Conc.	AQI	Raw Conc.
count	7899.000000	7899.000000	7899.000000
mean	29.241853	88.832257	32.177744
std	54.750358	64.285820	22.010910
min	-999.000000	-999.000000	0.000000
25%	19.300000	66.000000	19.000000
50%	26.100000	80.000000	26.000000
75%	40.000000	112.000000	40.000000
max	151.800000	202.000000	586.000000

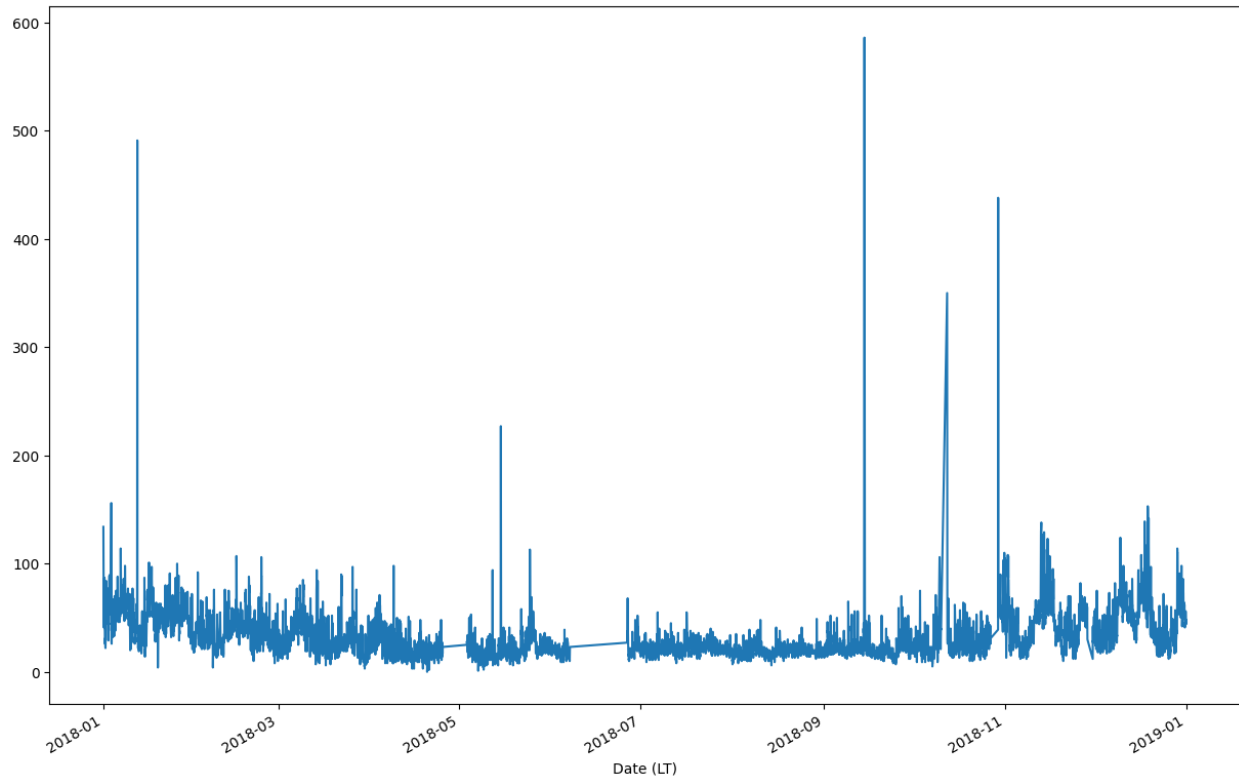
Screenshot_6



Screenshot_7



Followed steps of the coding screenshot is given here.



The Colombo 2018 air quality model indicates that there were notable variations in PM2.5 levels throughout the year. The PM2.5 particle concentration seems to have been high overall, occasionally above 500 $\mu\text{g}/\text{m}^3$.

Here's a more thorough explanation:

- 1.The date is shown on the x-axis, which runs from January 2018 to January 2019.
- 2.The concentration of PM2.5 particles is shown on the y-axis (probably measured in micrograms per cubic meter, or $\mu\text{g}/\text{m}^3$). Lower air quality is indicated by higher readings.
- 3.The average daily PM2.5 concentration for the whole year is displayed by the plotted line.

When analyzing this image, keep in mind a few limitations:

- 1.The graphic just displays daily average values. Daily variations that occur within a day are not recorded.
- 2.It is challenging to ascertain the precise concentration levels because the y-axis' scale is not stated.
- 3.Since the data's source is unknown, it's vital to exercise caution before making firm judgments.

Colombo's air quality represents a complex interplay of geographical, meteorological, and anthropogenic factors, necessitating a comprehensive approach to modeling and mitigation efforts. The city's urban landscape, characterized by dense vehicular traffic, industrial activities, construction dust, and biomass burning, poses significant challenges to air quality management. Vehicular emissions, in particular, emerge as a major contributor to pollution levels, exacerbated by rapid urbanization and population growth. Through the analysis of air quality data for Colombo in 2018, it becomes evident that understanding the spatial and temporal distribution of pollutant emissions is crucial for effective modeling and mitigation strategies. By leveraging advanced modeling techniques, policymakers and stakeholders can identify pollution hotspots, forecast air quality trends, and implement targeted interventions to mitigate the adverse impacts on public health and the environment.

Furthermore, the findings underscore the importance of integrating interdisciplinary approaches, incorporating data from meteorological observations, emission inventories, and socio-economic factors, to develop robust air quality models. Such models can serve as valuable tools for decision-makers to formulate evidence-based policies and interventions aimed at improving air quality and safeguarding the well-being of Colombo's residents. Moreover, fostering public awareness and participation is essential for promoting sustainable behaviors and fostering a culture of environmental stewardship in the community.